Christopher Douglas Robert Wyatt

Curriculum Vitae

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Profile

I started my scientific career studying phylogenetics and molecular evolution, which got me hooked on understanding how the complexity of life evolved. With the arrival of improved DNA/RNA sequencing technologies, this opened up a world of high quality empirical data to better understand molecular regulation and evolution.

I have had the privilege to work with most of these next-generation sequencing technologies (RNA-Seq, MS-Proteomics, Hi-C) to address key questions in the field of evolutionary and developmental biology. I am proficient in Unix, Perl and R, and have working experience with Python/Matlab. I am also proficient in writing higher-level workflows using Nextflow, which allows the sharing of reproducible, concise and scalable pipelines. I have presented my work at many major meetings across Europe and am confident communicating my ideas and science.

I completed my doctoral training at the Centre for Genomic Regulation (CRG, Barcelona), where I worked on alternative splicing regulation in the lab of Manuel Irimia and additionally collaborated with the labs of Peter Holland (Oxford), Nacho Maeso (CABD, Sevilla) and Bill Keyes (IGBMC, Strasbourg), working in the fields of transcriptional regulation, reprogramming, senescence and splicing regulation, producing a number of high impact publications and gaining a broad knowledge of many bioscience fields.

Since my PhD, I have worked at UCL (lab of Seirian Sumner) on several projects, including the use of machine learning to discover genes that are required for caste identification across wasps. More recently I have worked for Seqera Labs (the authors of Nextflow) to process biomedical data for specific startups, which has given me great experience in producing high quality informatics pipelines for clients.

Education & Research Career

Pipeline Developer

Seqera Labs, Barcelona *June 2021 – Sept 2021*

Writing Nextflow pipelines for several startups as well as documenting the training material for Nextflow. This gave me in-depth knowledge of this domain specific language for pipeline development.

Post-doctorate in Evolutionary Genomics / Transcriptomics

Centre for Biodiversity and Environment Research, UCL, London *April 2019 – Feb 2021*

Worked on a variety of research projects in the Eusocial Insect Research Group (Seirian Sumner) using machine learning, transcriptomics and proteomics approaches (using Nextflow) to explore wasp caste identity and evolution. Work submitted to bioRxiv prior to publication.

PhD in Biomedicine

Systems Biology, Centre for Genomic Regulation, Barcelona *Sept 2015 – March 2019*

Studied the regulation and patterns of alternative splicing during pre-implantation mammalian development.

Technical Skills

Languages & tools: Unix/Bash, Perl, R, Python, Matlab

Workflows: Nextflow (nf-core), reproducible & FAIR pipelines

Data types: RNA-Seq, MS-Proteomics, Hi-C, genome assembly